

GECON 7201

Applied Econometrics

Assignment 1
Due Thursday October 8, 2026

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Instructions

- Edit this Quarto document for your assignment. All answers must be typed in this document. Answers are to be written in the callout blocks (see below)
- Provide a link to your website in Nexus, so that I can view your assignment.
- Upload both a PDF and Quarto file to the nexus assignment portal.
- The use of AI is strictly prohibited for this assignment.
- Show your work
- Type your mathematical answers using LaTeX
- For questions marked R, include both your code and its output.
- You may use `solve()`, `t()`, `%*%`, `diag()`, `qr()`, `eigen()`, and other base R functions freely unless a question says otherwise.
- Please round numerical answers to two decimal places unless stated otherwise.

```
set.seed(3201)
```

Questions

Q1

Create a personal Quarto website hosted on GitHub pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

- a. An “Index” or “Home” page. This page should include the following: - Your name - Your photo (optional) - Your credentials - An “About Me” section that provides a brief (one paragraph) introduction to yourself. [10 Marks]

- b. A blog page. This will be the landing site for future assignments [5 Marks]
- c. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
- d. Navigation bar with “Home” and “Blog” buttons. The navigation bar is edited in the “_quarto.yml” file. [2 Marks]

Paste your link here

Q2

Consider

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & -3 \\ 0 & -3 & 6 \end{bmatrix}.$$

- a. State the dimensions of A and identify a_{23} . [2 Marks]

The dimensions of A are 3x3. The element a_{23} is -1

- b. Is A a square matrix? Is it symmetric? Justify your answer using the definition $A = A^\top$, not just by inspection. [2 Marks]

The matrix A is a square matrix because it has 3 rows and 3 columns. A is symmetric. To check whether A is a symmetric, we compare A with its transpose.

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}, \quad A^\top = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$$

Since $A = A^\top$, A is symmetric

- c. Is A a diagonal matrix? Explain why or why not. [1 Mark]

A diagonal matrix must have zeros everywhere except on the main diagonal. Here, $A_{13} = 1$, $a_{31} = 1$, $a_{23} = -1$, $a_{32} = -1$ Therefore, it is not a diagonal.

- d. In R, Enter A and B into R. Use `isSymmetric()` to confirm your answer to part (b) for both matrices. [4 Marks]

```

A <- matrix(c(2, 0, 1,
              0, 3, -1,
              1, -1, 4),
            nrow = 3, byrow = TRUE)

B <- matrix(c(1, 2, 0,
              2, 5, -3,
              0, -3, 6),
            nrow = 3, byrow = TRUE)

isSymmetric(A)
isSymmetric(B)

```

Q3

Using the same A and B from Question 1:

- a. Compute $A + B$ and $3A - 2B$ by hand. [4 marks]

$$\begin{aligned}
 A + B &= A \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & -3 \\ 0 & -3 & 6 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 8 & -4 \\ 1 & -4 & 10 \end{bmatrix}. \\
 3A - 2B &= 3 \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix} - 2 \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & -3 \\ 0 & -3 & 6 \end{bmatrix} \\
 &= \begin{bmatrix} 6 & 0 & 3 \\ 0 & 9 & -3 \\ 3 & -3 & 12 \end{bmatrix} - \begin{bmatrix} 2 & 4 & 0 \\ 4 & 10 & -6 \\ 0 & -6 & 12 \end{bmatrix} = \begin{bmatrix} 4 & -4 & 3 \\ -4 & -1 & 3 \\ 3 & 3 & 0 \end{bmatrix}
 \end{aligned}$$

- b. Compute AB by hand, showing at least one entry's calculation in full using the summation formula $(AB)_{ij} = \sum_{\ell} a_{i\ell}b_{\ell j}$. [2 marks]

Using the summation formula, $(AB)_{ij} = \sum_{\ell} a_{i\ell}b_{\ell j}$.

$$AB = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & -3 \\ 0 & -3 & 6 \end{bmatrix}$$

For the top left-entry, using the summation formula,

$$2(1) + 0(2) + 1(0) = 2$$

$$AB = \begin{bmatrix} 2 & 1 & 6 \\ 6 & 18 & -15 \\ -1 & -15 & 27 \end{bmatrix}.$$

- c. Compute BA . Is $AB = BA$ here? Was this guaranteed in general? [2 marks]

Tip

$$BA = \begin{bmatrix} 2 & 6 & -1 \\ 6 & 18 & -15 \\ -1 & -15 & 27 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 & 1 & 6 \\ 6 & 18 & -15 \\ -1 & -15 & 27 \end{bmatrix}$$

Here, AB and BA are not equal because the matrices are different. Mo they are not guaranteed in general.

- d. In R, verify parts (a)–(c) in R. Also compute $A * B$ (element-wise) and explain in one sentence why this is a different operation from $A \%*\% B$. [4 marks]

Tip

```
A + B
3*A - 2*B
A \%*\% B
B \%*\% A
A * B
```

‘ $A * B$ ’ is different from $A \%*\% B$ *because* multiples corresponding elements but $\%*\%$ multiplies rows and columns.

Q4

Let

$$X = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

- a. Write down $X^\top$ and state its dimensions. [2 marks]

Tip

$$x^\top = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$x^\top$ has a 2x3 dimensions.

- b. Compute $X^\top X$ and $XX^\top$ by hand. Explain why these two products have different dimensions even though both involve X and $X^\top$. [4 marks]

Tip

$$x^\top = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}$$

$$xx^\top = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix}$$

The two products have different dimensions because the order in which we multiply the matrices is different.

- c. Let $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Compute $\mathbf{v}^\top \mathbf{v}$ and $\mathbf{v}\mathbf{v}^\top$ by hand, and state the dimension of each result. [4 marks]

Tip

$$v = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

For $(\mathbf{v}^\top \mathbf{v})$, we multiply the transpose of $(\mathbf{v})$ by $(\mathbf{v})$:

$$v^T v = \begin{bmatrix} 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = 1(1) + (-1)(-1) + 2(2) = 6$$

Therefore, $(\mathbf{v}^\top \mathbf{v})$ is a (1×1) matrix (a scalar). For $(\mathbf{v}\mathbf{v}^\top)$:

$$vv^T = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix}$$

- d. In R, verify parts (a)–(c). Then confirm numerically, for two conformable matrices of your choosing, that $(AB)^{\top} = B^{\top}A^{\top}$. [4 marks]

Tip

```
X <- matrix(c(1,4,2,5,3,6), nrow = 3, byrow = TRUE)

v <- matrix(c(1,-1, 2), nrow = 3)

t(X)
t(X) %*% X
X %*% t(X)

t(v) %*% v
v %*% t(v)

A <- matrix(c(1,2,3,4), nrow = 2, byrow = TRUE)
B <- matrix(c(5,6,7,8), nrow = 2, byrow = TRUE)

t(A %*% B)
t(B) %*% t(A)

all.equal(t(A %*% B), t(B) %*% t(A))
```

This analysis verifies the calculations from (a)–(c). The output for $t(X)$, $t(X)X$, $Xt(X)$, $t(v)v$, and $vt(v)$ matches the results calculated by hand.

Q5

Consider

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}.$$

- a. In R, compute A^{-1} using `solve()`. [2 marks]

Tip

```
A <- matrix(c(4, 2,
              3, 3),
            nrow = 2, byrow = TRUE)

solve(A)
```

- b. Construct a 3×3 matrix that is singular (for example, make one column a linear combination of the others) and show what happens when you call `solve()` on it. [2 marks]

Tip

```
S <-matrix (c(1, 2, 3,
              2, 4, 6,
              3, 6, 9),
            nrow = 3, byrow = TRUE)
solve (S)
```

Q6

- a. Define, in your own words, what it means for a set of vectors to be linearly independent. [1 mark]

Tip

A set of vectors is linearly independent if the only way to form a zero vector is for all the coefficients to be zero.

- b. Consider

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}.$$

Show that $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are linearly dependent by finding constants c_1, c_2, c_3 , not all zero, such that $c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3 = \mathbf{0}$. [3 marks]

Tip

$$\begin{aligned} c_1x_1 + c_2x_2 + c_3x_3 &= 0 \\ c_1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + c_3 \begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ c_1 + 2c_3 &= 0 \\ 2c_1 + c_2 + 5c_3 &= 0 \\ c_1 + c_2 + 3c_3 &= 0 \end{aligned}$$

Choose $c_3 = 1$. Then:

$$c_1 = -2, \quad c_2 = -1$$

Therefore,

$$c_1 = -2, \quad c_2 = -1, \quad c_3 = 1$$

and

$$-2x_1 - x_2 + x_3 = 0.$$

Hence, the vectors are linearly dependent.

- c. Let $M = [\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{x}_3]$. What is $\text{rank}(M)$? Explain your reasoning using your answer to part (b). [2 marks]

Tip

Since x_1 , x_2 , and x_3 are linearly dependent, the three columns of M cannot all be linearly independent.

- d. In R, build M and confirm its rank using `qr(M)$rank`. Then modify one entry of $\mathbf{x}_3$ so that the three vectors become linearly independent, and confirm the new rank. [3 marks]

Tip

```
x1 <- c(1, 2, 1)
x2 <- c(0, 1, 1)
x3 <- c(2, 5, 3)

M <- cbind(x1, x2, x3)

M
qr(M)$rank
```

Q7.

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

- a. Write out the quadratic form $\mathbf{x}^\top A \mathbf{x}$ in terms of x_1 and x_2 . [2 marks]

Tip

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$
$$x^\top A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Firstly,

$$Ax = \begin{bmatrix} 2x_1 + x_2 \\ x_1 + 2x_2 \end{bmatrix}.$$

Therefore,

$$x^\top Ax = 2x_1^2 + 2x_1x_2 + 2x_2^2.$$

- b. Is B invertible? Justify your answer using the relationship between definiteness, rank, and invertibility discussed in class. [3 marks]

Tip

B is not invertible because its two columns are identical. Therefore, the columns are linearly dependent and B is singular and does not have an inverse.

- c. In R, Confirm parts (a) and (b) by computing `eigen(A)$values` and `eigen(B)$values`. State the rule connecting the sign of a symmetric matrix's eigenvalues to whether it is positive definite, positive semidefinite, or neither. [4 marks]

Tip

```
A <- matrix(c(2, 1,
              1, 2),
            nrow = 2, byrow = TRUE)

B <- matrix(c(1, 1,
              1, 1),
            nrow = 2, byrow = TRUE)

eigen(A)$values
eigen(B)$values
```

Q8

Let

$$A = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}.$$

- a. Show by direct calculation that $A^2 = A$, i.e., that A is idempotent. [2 marks]

Tip

$$A^2 = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.25 + 0.25 & 0.25 + 0.25 \\ 0.25 + 0.25 & 0.25 + 0.25 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} = A.$$

$$\boxed{A^2 = A}$$

A is idempotent.

- b. Using only the idempotent property $A^2 = A$ (not this specific matrix), show that $I - A$ is also idempotent. [2 marks]

Tip

$$(I - A)^2 = (I - A)(I - A)$$

$$= I - 2A + A^2$$

$$= I - 2A + A$$

$$I - A.$$

$I - A$ is also idempotent.

- c. In R, confirm that A is idempotent using matrix multiplication in R. [1 mark]

```
A <- matrix(c(0.5, 0.5,
              0.5, 0.5),
            nrow = 2, byrow = TRUE)
A %% A
```

Q9

Let $\mathbf{a} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and

$$S = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Note that S is symmetric.

- a. Write down $\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}}$ for a general 3×1 vector $\mathbf{x}$. (No calculation needed, state the rule and apply it to this $\mathbf{a}$.) [2 marks]

Tip

$$\frac{\text{partial}(a^\top x)}{\partial x} = a$$

$$a = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$$

- b. Derive $\frac{\partial(\mathbf{x}^\top S \mathbf{x})}{\partial \mathbf{x}}$ using the general rule for $\frac{\partial(\mathbf{x}^\top A \mathbf{x})}{\partial \mathbf{x}}$, and simplify using the fact that S is symmetric. [2 marks]

Tip

$$\frac{\text{partial}(a^\top A x)}{\partial x} = (A + A^\top)x.$$

$$\frac{\text{partial}(a^\top S x)}{\partial x} = (S + S^\top)x.$$

Since S is symmetric, $S^\top = S$. Because of that,

$$\frac{\text{partial}(a^\top S x)}{\partial x} = (S + S)x = 2Sx.$$

- c. Evaluate your answer to part (b) at $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$. [1 mark]

Tip

$$\text{At } \mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$2Sx = 2 \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}.$$

- d. In R, using the `numDeriv` package (`grad()`), numerically confirm your answers to parts (a) and (c). [2 marks]

```

library(numDeriv)

# Part(a)
a <- c(3, -2, 1)
f <- function(x) {
  sum(a * x)
}
grad(f, c(1, 0, 0))

# Part (c)
S <- matrix(c(2, 1, 0,
              1, 3, 1,
              0, 1, 2),
            nrow = 3, byrow = TRUE)

g <- function(x) {
  as.numeric(t(x) %*% S %*% x)
}
grad(g, c(1, 0, -1))

```

Q10

Let $\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$ be a random vector with

$$\mu = E[\mathbf{X}] = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \quad \Sigma = \text{Var}(\mathbf{X}) = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}.$$

Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$, and define $\mathbf{Y} = A\mathbf{X} + \mathbf{b}$.

a. State $E[\mathbf{Y}]$ and $\text{Var}(\mathbf{Y})$, citing the properties used. [3 marks]

a.

$$E[Y] = A\mu + b$$

and

$$\text{Var}(Y) = A\Sigma A^T.$$

b. Compute $E[\mathbf{Y}]$ by hand. [2 marks]

Tip

$$\begin{aligned} & E[\mathbf{Y}] \\ = & \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 5 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 5 \\ -2 \end{bmatrix} = \boxed{\begin{bmatrix} 7 \\ -1 \end{bmatrix}} \end{aligned}$$

- c. Compute $\text{Var}(\mathbf{Y}) = A\Sigma A^\top$ by hand, showing your intermediate matrix products. [3 marks]

Tip

$$\begin{aligned} & \begin{bmatrix} 5 & 3 & -1 \\ 1 & 3 & -4 \end{bmatrix} \quad A^\top = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & -1 \end{bmatrix} \\ & A\Sigma A^\top = \begin{bmatrix} 5 & 3 & -1 \\ 1 & 3 & -4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & -1 \end{bmatrix} = \boxed{\begin{bmatrix} 8 & 6 \\ 6 & 7 \end{bmatrix}} \end{aligned}$$

- d. Let $\mathbf{a} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Compute $\text{Var}(\mathbf{a}^\top \mathbf{X})$ using the appropriate property, without recomputing $\text{Var}(\mathbf{X})$ from scratch. [3 marks]

Tip

$$\text{Var}(\mathbf{a}^\top \Sigma \mathbf{a}) = 20.$$

- e. In R, using `MASS::mvrnorm()`, draw 100,000 observations from a distribution with mean μ and variance Σ (you may treat $\mathbf{X}$ as multivariate normal for this simulation). Confirm your answers to (b), (c), and (d) using `colMeans()` and `cov()`. [3 marks]

Tip

```
library(MASS)

mu <- c(2, 0, 01)
Sigma <- matrix (c(4, 1, 0,
                  1, 2, -1,
                  0, -1, 3),
                nrow = 3, byrow = TRUE)
x <- mvrnorm(100000, mu = mu, Sigma = Sigma)

colMeans(x)
cov(x)
```

Q11

- a. Suppose $\mathbf{X} \sim N_3(\mu, \Sigma)$ using the μ and Σ from Question 10. What is the marginal distribution of X_2 alone? Give its distribution and parameters. [2 marks]

Tip

$$x_2 \sim N(0, 2)$$

- c. Using property 4 of the multivariate normal (zero covariance implies independence), determine whether X_1 and X_3 are independent. Justify your answer using an entry of Σ . [1 mark]

Tip

Your answer here

- d. In R, using your simulated draws from Question 10(e), produce a scatterplot of X_1 against X_3 and compute `cor(X1, X3)`. Comment on whether the plot is consistent with your answer to part (c). [3 marks]

Tip

```
x1 <- as.numeric(x[,1])
x3 <- as.numeric(x[,3])

plot (x1, x3, xlab = "x1", ylab = "x3")
cor(x1, x3)
```